

File Inventory — EMS Readiness Optimization Project

Generated: 2026-03-12 | Version: 1.2.0 | Total tracked files in remote: **210+** | Total local files: **220+**

Phase Deliverable Checklist

Phase	Description	Status
Phase 1	Data Processing & EDA	✓ Complete
Phase 2	Demand & Service Modeling (NHPP)	✓ Complete
Phase 3	Optimization Models (P0, P1, P2)	✓ Complete
Phase 4	DES Simulation with V&V (39 tests)	✓ Complete
Phase 5	Experimental Design & Production Runs (1,440 sims)	✓ Complete
Phase 6	Comprehensive Statistical Analysis	✓ Complete
Phase 7	Final Report & Documentation	✓ Complete
Phase 8	Alternative Analyses & Extensions	✓ Complete

Directory Structure & File Inventory

configs/ — Configuration Files

File	Size	Description
demand.yaml	856 B	NHPP demand model parameters
optimization.yaml	960 B	MIP solver and unit-count settings
service.yaml	1.5 KB	Travel time and service time config
simulation.yaml	1.1 KB	DES engine and batch runner settings

data/raw/ — Raw Input Data

File	Size	Tracked	Description
FDNY_Firehouse_Listing_20260223.csv	34 KB	✓	219 FDNY firehouses
FDNY_Firehouse_Listing_Data_Dictionary.xlsx	66 KB	✓	Firehouse field definitions
Motor_Vehicle_Collisions_-_Crashes_20260223.csv	536 MB	✗	2.24M crash records (too large)
Motor_Vehicle_Collisions_Data_Dictionary.xlsx	96 KB	✓	Crash field definitions
Precincts_20260223.csv	3.6 MB	✗	78 precinct boundaries (large)
Precincts_Data_Dictionary.xlsx	62 KB	✓	Precinct field definitions
manhattan_boundary.geojson	247 KB	✓	Manhattan polygon boundary
cbd_boundary.geojson	101 KB	✓	CBD/Congestion Relief Zone boundary
nyc_borough_boundaries.geojson	3.1 MB	✓	All 5 NYC borough boundaries

data/processed/ — Processed Data

File	Size	Tracked	Description
crashes_manhattan.csv	99 MB	✗	Filtered Manhattan crashes (large)
crashes_manhattan.parquet	15 MB	✗	Parquet version of above
de-mand_lambda_hourly.csv	1.9 KB	✓	Hourly λ factors (24 hours)
de-mand_lambda_dow.csv	653 B	✓	Day-of-week λ factors
de-mand_lambda_precinct.csv	1.7 KB	✓	Precinct-level λ rates
de-mand_model_summary.json	696 B	✓	Demand model summary stats
distance_matrix_firehouse_precinct.csv	27 KB	✓	48×30 Haversine distance matrix
distance_matrix_firehouse_precinct_manhattan.csv	27 KB	✓	48×30 Manhattan distance matrix
fire-houses_clean.csv	32 KB	✓	All cleaned firehouses
fire-houses_manhattan.csv	7.1 KB	✓	48 Manhattan firehouses
pre-cincts_manhattan.geojson	640 KB	✓	30 Manhattan precincts

src/ems_readiness/ — Source Code (7,800+ lines total)

Module	Files	Description
demand/	<code>__init__.py</code> , <code>arrival_generator.py</code>	NHPP arrival generation (thinning algorithm)
service/	<code>__init__.py</code> , <code>travel_time.py</code> , <code>service_time.py</code>	Travel proxy & LogNormal service time
optimization/	<code>__init__.py</code> , <code>models.py</code> , <code>policies.py</code> , <code>allocator.py</code>	MIP formulations (incl. CBD-focused) & baseline policies
simulation/	<code>__init__.py</code> , <code>engine.py</code> , <code>entities.py</code> , <code>resources.py</code> , <code>dispatcher.py</code> , <code>metrics.py</code> , <code>runner.py</code>	SimPy DES engine with batch runner
utils/	<code>__init__.py</code> , <code>distance.py</code>	Haversine & Manhattan distance calculations

scripts/ — Automation Scripts

File	Description
<code>data_audit.py</code>	Data quality audit pipeline
<code>audit_step1_boundaries.py</code>	Boundary file validation
<code>audit_step2_firehouses.py</code>	Firehouse data validation
<code>audit_step3_precincts.py</code>	Precinct data validation
<code>audit_step4_crashes.py</code>	Crash data validation
<code>demand_modeling.py</code>	NHPP demand model fitting
<code>run_optimization_comparison.py</code>	Multi-model optimization comparison
<code>run_verification.py</code>	Simulation verification tests
<code>run_validation_pilots.py</code>	Validation pilot experiments
<code>run_production_experiments.py</code>	Full 1,440-run production experiments
<code>analyze_production_results.py</code>	Production result analysis
<code>generate_publication_figures.py</code>	Publication-quality figure generation
<code>generate_summary_dashboard.py</code>	Project summary dashboard
<code>run_cbd_experiment.py</code>	CBD robustness experiment (330 runs)
<code>analyze_queue_metrics.py</code>	Queue metrics analysis
<code>analyze_seasonal_patterns.py</code>	Seasonal variation analysis
<code>generate_manhattan_distance_matrix.py</code>	Manhattan (taxicab) distance matrix generation
<code>run_distance_comparison_experiment.py</code>	Haversine vs. Manhattan metric comparison
<code>run_cbd_focused_optimization.py</code>	CBD-focused vs. Manhattan-wide optimization

tests/ — Test Suite (39 tests)

File	Tests	Description
<code>conftest.py</code>	—	Shared fixtures
<code>test_simulation_core.py</code>	15	Core DES engine tests
<code>test_dispatch_logic.py</code>	10	Dispatcher & resource tests
<code>test_extreme_cases.py</code>	8	Edge case & stress tests
<code>test_reproducibility.py</code>	6	Seeded reproducibility tests

notebooks/ — Analysis Notebooks (7 notebooks)

File	Description
<code>02_eda_spatiotemporal.ipynb</code>	Exploratory data analysis
<code>03_input_modeling.ipynb</code>	NHPP demand model fitting
<code>04_service_travel_proxy.ipynb</code>	Service & travel time validation
<code>05_optimization.ipynb</code>	Optimization model comparison
<code>06_simulation_debug.ipynb</code>	Simulation debugging & V&V
<code>07_production_results.ipynb</code>	Production experiment results
<code>08_statistical_analysis.ipynb</code>	Comprehensive statistical analysis

results/figures/ — Visualizations (33 PNG files)

Category	Files	Description
EDA	10	Crash heatmap, temporal trends, firehouses map, etc.
Optimization	4	Inputs, sensitivity, allocation comparison
Simulation V&V	4	Verification timeline, validation comparisons
Experiments	4	Exp1-Exp4 result plots
Publication	5	pub_fig1-pub_fig5 (high quality)
Model	6	Distance heatmap, travel time, service time, NHPP

results/tables/ — Statistical Tables (18 files)

File	Description
descriptive_statistics.csv	Summary stats for all experiments
anova_results.csv	ANOVA test results
posthoc_comparisons.csv	Pairwise comparisons (Tukey HSD)
confidence_intervals.csv	95% confidence intervals
effect_sizes.csv	Cohen's d effect sizes
sensitivity_summary.csv	Sensitivity analysis summary
table1-table4_*.csv/.tex	Publication-ready tables (CSV + LaTeX)
exp1-exp4_*.csv	Per-experiment pivot tables
optimization_comparison.csv	Policy comparison metrics

results/simulation/ — Simulation Outputs

Directory	Files	Description
production/	6	4 experiment CSVs + summary + log
validation_pilot/	5	3 pilot JSONs + 2 comparison tables
verification/	4	4 verification test JSONs

results/optimization/ — Optimization Results

File	Description
allocations_K20–K48.csv	Unit allocations for K=20,30,40,48
policy_comparison.csv	5-policy comparison results
sensitivity_analysis.csv	K-sensitivity analysis
findings_summary.json	Machine-readable summary
PHASE3_SUMMARY.md/.pdf	Phase 3 narrative summary

results/distance_comparison/ — Distance Metric Comparison (Phase 8)

File	Description
comparison_table.csv	Haversine vs. Manhattan simulation metrics
allocation_comparison.csv	Side-by-side firehouse allocations
distance_matrices_heatmap.png	Dual-panel heatmap
distance_scatter.png	Manhattan vs. Haversine scatter
distance_comparison_bar.png	RT comparison bar chart
distance_comparison_boxplot.png	RT distribution boxplot
experiment_log.txt	Experiment execution log

results/cbd_focused_comparison/ — CBD-Focused Optimization (Phase 8)

File	Description
comparison_table.csv	CBD-focused vs. Manhattan-wide metrics
allocations.csv	Side-by-side allocations
cbd_focused_comparison.png	RT comparison bar chart
allocation_comparison.png	Allocation distribution comparison
equity_tradeoff.png	CBD vs. non-CBD equity trade-off
experiment_log.txt	Experiment execution log

docs/ — Documentation (30+ files)

File	Description
project_charter.md	Project scope and objectives
source_manifest.md/.pdf	Data source inventory
assumptions_log.md	Documented assumptions
decisions_log.md/.pdf	Key decision records
blocker_log.md	Issue tracking
demand_model_spec.md/.pdf	NHPP demand model specification
service_model_spec.md/.pdf	Travel & service model spec
optimization_formulation.md/.pdf	MIP formulation documentation
optimization_results.md/.pdf	Optimization results narrative
conceptual_model.md/.pdf	DES conceptual model
experimental_design.md/.pdf	Factorial experimental design
verification_log.md/.pdf	V&V test results
output_analysis.md/.pdf	Statistical analysis report
executive_summary.md/.pdf	Executive summary
technical_report.md	Comprehensive final report
executive_presentation.md	Slide deck for stakeholders
implementation_roadmap.md	Deployment plan
project_archive.md	Project archive & timeline
code_documentation.md	Code architecture guide
file_inventory.md	This file
cbd_robustness_analysis.md	CBD robustness DES experiment
queue_analysis.md	Queueing performance analysis
gap_closure_report.md	Gap closure verification
distance_metric_comparison.md/.pdf	Haversine vs. Manhattan comparison (Phase 8)

File	Description
<code>cbd_focused_optimization_analysis.md/.pdf</code>	CBD-focused optimization report (Phase 8)
<code>alternative_analyses_summary.md/.pdf</code>	Alternative analyses summary (Phase 8)
<code>research_questions_assessment.md</code>	Research questions assessment with simulation evidence
<code>project_workflow_wbs.md/.pdf</code>	Work Breakdown Structure

Project Statistics

Metric	Value
Total Python LOC	7,800+
Total source modules	14
Total scripts	19
Total test cases	39
Total notebooks	8
Total figures	51+
Total CSV result files	30+
Total documentation files	30+
Total simulation runs	1,770+ (production) + alternative analyses
Tracked files in Git	210+
Project size (excl. raw data)	~140 MB